



VIDYARTHI VIGYAN MANTHAN 2026-27

SYLLABUS (CLASS - XI)

CLASS XI

PHYSICS

- 1. Units and Measurements:** Need for measurement: Units of measurement; systems of units; SI units, fundamental and derived units. significant figures, Determining the uncertainty in the result. Dimensions of physical quantities, dimensional analysis and its applications.
- 2. Kinematics:** Motion in a Straight Line Frame of reference, Motion in a straight line, Elementary concepts of differentiation and integration for describing motion, uniform and non-uniform motion, average speed and average velocity and instantaneous velocity, uniformly accelerated motion, velocity-time and position-time graphs. Relations for uniformly accelerated motion (graphical and calculus treatment).
- 3. Motion in a Plane:** Scalar and vector quantities; position and displacement vectors, general vectors and their notations; equality of vectors, multiplication of vectors by a real number; addition and subtraction of vectors, Unit vector; resolution of a vector in a plane, rectangular components, Scalar and Vector product of vectors. Motion in a plane, cases of uniform velocity and uniform acceleration- projectile motion, uniform circular motion.
- 4. Laws of Motion:** Laws of Motion. Intuitive concept of force, Inertia, Newton's first law of motion, momentum, Newton's second law of motion, impulse, Newton's third law of motion. Law of conservation of linear momentum and its applications. Equilibrium of concurrent forces, Static and kinetic friction, laws of friction, rolling friction, and lubrication. Dynamics of uniform circular motion: Centripetal force, examples of circular motion (vehicle on a level circular road, vehicle on a banked road).
- 5. Work, Energy and Power:** Work, Energy and Power. Work done by a constant force and a variable force; kinetic energy, work-energy theorem, and power. Notion of potential energy, potential energy of a spring, conservative forces, non-conservative forces, motion in a vertical circle; elastic and inelastic collisions in one and two dimensions.
- 6. Motion of System of Particles and Rigid Body:** System of Particles and Rotational Motion Centre of mass of a two-particle system, momentum conservation and Centre of mass motion. Centre of mass of a rigid body; centre of mass of a uniform rod. Moment of a force, torque, angular momentum, law of conservation of angular momentum and its applications. Equilibrium of rigid bodies, rigid body rotation and equations of rotational motion, comparison of linear and rotational motions. Moment of inertia, radius of gyration, values of moments of inertia for simple geometrical objects (no derivation).
- 7. Gravitation:** Kepler's laws of planetary motion, universal law of gravitation. Acceleration due to gravity and its variation with altitude and depth. Gravitational potential energy, gravitational potential, escape speed, orbital velocity of a satellite, and energy of an orbiting satellite.



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8. **Properties of Bulk Matter:** Mechanical Properties of Solids Elasticity, Stress-strain relationship, Hooke's law, Young's modulus, bulk modulus, shear modulus of rigidity (qualitative idea only), Poisson's ratio; elastic energy. Application of the elastic behaviour of materials (qualitative idea only).
9. **Mechanical Properties of Fluids:** Pressure due to a fluid column; Pascal's law and its applications (hydraulic lift and hydraulic brakes), effect of gravity on fluid pressure. Viscosity, Stokes' law, terminal velocity, streamline and turbulent flow, critical velocity, Bernoulli's theorem and its simple applications (Torricelli's law and Dynamic lift). Surface energy and surface tension, angle of contact, excess of pressure across a curved surface, application of surface tension ideas to drops, bubbles and capillary rise.
10. **Thermal Properties of Matter:** Heat, temperature, thermal expansion; thermal expansion of solids, liquids and gases, anomalous expansion of water; specific heat capacity; C_p , C_v - calorimetry; change of state - latent heat capacity. Heat transfer- conduction, convection and radiation, thermal conductivity, qualitative ideas of Blackbody radiation, Wein's displacement Law, Stefan's law.
11. **Thermodynamics:** Thermodynamics, Thermal equilibrium and definition of temperature, zeroth law of thermodynamics, heat, work and internal energy. First law of thermodynamics, Second law of thermodynamics: Thermodynamic state variable and equation of state. Change of condition of gaseous state isothermal, adiabatic, reversible, irreversible, and cyclic processes.
12. **Behavior of Perfect Gases and Kinetic Theory of Gases:** Kinetic Theory Equation of state of a perfect gas, work done in compressing a gas. Kinetic theory of gases - assumptions, concept of pressure. Kinetic interpretation of temperature; rms speed of gas molecules; degrees of freedom, law of equi-partition of energy (statement only) and application to specific heat capacities of gases; concept of mean free path, Avogadro's number.
13. **Oscillations and Waves:** Oscillations, Periodic motion - time period, frequency, displacement as a function of time, periodic functions and their applications. Simple harmonic motion (S.H.M), uniform circular motion and its equations of motion; phase; oscillations of a loaded spring- restoring force and force constant; energy in S.H.M. Kinetic and potential energies; simple pendulum derivation of expression for its time period. Wave motion: Transverse and longitudinal waves, speed of travelling wave, displacement relation for a progressive wave, principle of superposition of waves, reflection of waves, standing waves in strings and organ pipes, fundamental mode and harmonics, Beats.

CHEMISTRY

- Some Basic Concepts of Chemistry
- Structure of Atoms
- Classification of Elements and Periodicity in properties
- Chemical Bonding and Molecular Structure
- Thermodynamics



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- Equilibrium
- Redox Reaction
- Organic Chemistry: Some Basic Principles and Techniques
- Hydrocarbons

BIOLOGY

- Living world diversity and biological classification
- Plant and animal kingdom
- Morphology and anatomy of flowering plants
- Structural organization in animals
- Structure of the cell, cell division
- Biomolecules
- Photosynthesis, respiration, growth and development in plants
- Respiration, circulation, excretion, nervous and endocrine system in animals

MATHEMATICS

1. **Number Theory:** Sets – Representation, Types, Subsets, Operations on Sets, Applications
Relations – Cartesian Product, Relations, Domain, Co-domain, Range, Arrow Diagram,
Complex Numbers – Operations, Argand Plane, Modulus and Argument, Polar Form, De
Moivre's Theorem, Logarithms – Definition and Results.
2. **Algebra:** Linear Inequalities – In one and Two Variables, Quadratic Equation, Inequalities,
Wavy Curve Method Binomial Theorem – Expansion, General & Middle Term, Sequences
& Series – GP, HP, AM, GM, Special Series.
3. **Trigonometry:** Trigonometric Equations and Inequations, Properties of Triangles, Inverse
Trigonometric Functions, Graphs.
4. **Geometry:** Straight Lines – Various Forms of a Line, Pair of Straight Lines, Conic Sections
– Circles, Parabola, Ellipse, Hyperbola Vectors – Components of a Vector, Dot and Cross
Product Three-Dimensional Geometry – Distance Formula & Section Formula.
5. **Combinatorics:** Counting Principles – Factorial, Fundamental Principles, Permutations,
Combinations, Recurrence Relations – Fibonacci Sequence.
6. **Calculus:** Functions - Domain, Co-domain, Range, Types – Graph, Limits – Definition,
Special Limits, Differentiation (Product, Quotient and Chain Rule), Integration based on
Substitution method and basic identity.
7. **Statistics & Probability:** Variance, Coefficient of Variation, Probability – Events, Types,
Algebra of Events, Probability, Addition and Multiplication Theorem of Probability,
Binomial and Poisson Distribution, Measure of Dispersions – Mean Deviation, Standard
Deviation.



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LOGIC AND REASONING

Pattern Recognition: Number Series, Alphabet Series, Image Series, Odd One Out, Analogy, Figure Matrix & Matrix Reasoning, Pattern-based Puzzles, Coding-Decoding, Logical Sequence of Words, meaningful word formation.

Analytical Reasoning: Blood Relations, Direction Sense, Order & Ranking, Seating Arrangement, Grid Puzzles, Games & Tournaments, Scheduling Problems, Clocks & Calendar, Sequencing & Ordering Problems, Sudoku.

Visual Reasoning: Mirror Images, Water Images, Rotation, Paper Folding, Paper Cutting, Figure Counting, Embedded Figures, Shape Construction, Spatial Visualization, 3D visualization, Cubes, Dice & Cube Nets, Best Fit, Symmetry.

Mathematical Reasoning: Mathematical Operations, Number Puzzles, Cryptarithmic (Alphametics) puzzles, Binary Logic, Logical Inequalities, Logical Venn Diagrams, Word Problems.

Critical Thinking: Data Sufficiency, Decision Making, Logical Connectives, Syllogism, Input Output series, Critical Path Analysis, Scheduling Problems.

Emerging Reasoning Skills: Constraint Satisfaction Problems, Follow the Algorithm, Algorithm Output Prediction, Scientific Reasoning, Logical Puzzle Sets, Optimization, Geometrical reasoning.

LIFE STORY OF INDIAN SCIENTIST(S)

The Trailblazing Journey of Rohini Godbole (All Chapters)

INDIAN CONTRIBUTIONS TO SCIENCE

Indian Contributions to Science : Volume-III